

SHAHIL SHAIK

Ph.D. Candidate in Mechanical Engineering | Robotics and Intelligent Systems

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RESEARCH PROFILE

Ph.D. candidate and Graduate Research Assistant developing efficient learning architectures for embodied and multi-agent intelligence. Research spans transformer attention, vision-language and vision-language-action models, multi-agent deep reinforcement learning, distributional policy optimization, and shared human-robot control. Published in the American Control Conference and *IET Cyber-Systems and Robotics*; experienced in reproducible, GPU-accelerated research with PyTorch.

EDUCATION

2021–Present

Ph.D. in Mechanical Engineering (Robotics)

Clemson University, Clemson, South Carolina

Expected graduation: August 2027. Research focus: intelligent multi-agent robotic systems and learning-based autonomy.

2019–2021

Graduate Study in Mechanical Engineering

Clemson University, Clemson, South Carolina

Entered the M.S. program in 2019 and transitioned to the Ph.D. program in 2021.

2015–2019

Bachelor's Degree in Mechanical Engineering

Anil Neerukonda Institute of Technology and Sciences, Visakhapatnam, India

GPA: 8.55/10.

ACADEMIC APPOINTMENT

Current

Graduate Research Assistant, Department of Mechanical Engineering

Clemson University, Clemson, South Carolina

Faculty advisor: Dr. Yue Wang.

RESEARCH INTERESTS

Efficient transformer architectures • Vision-language-action learning • Multi-agent reinforcement learning • Robot learning • Human-robot shared control • Distributed and memory-efficient deep learning

SELECTED RESEARCH

Lie-Generated Metric Attention (LGMA) | *Lead developer and investigator*

Developing a PyTorch attention architecture that generates head-specific metrics from shared Lie-algebra generators to reduce parameter and key-value cache costs. Built controlled MHA-versus-LGMA experiments, stability diagnostics, checkpointing, efficiency accounting, and single-/multi-GPU training workflows for TinyStories and robot-learning studies.

Vision-Language(-Action) Models for Multi-Agent Robotics | *Research and software development*

Investigating multimodal critics and action models for policy evaluation and robot learning, including MA-VLCM and planned LIBERO benchmarking. Work integrates visual observations, language-conditioned reasoning, temporal context, and multi-agent state representations.

Multi-Agent Deep Reinforcement Learning | *Doctoral research*

Developed methods for learning under constrained inter-agent communication and studied distributional policy-gradient estimation. Research emphasizes coordination, robustness, sample efficiency, and principled evaluation across multi-agent settings.

Shared Control of Mobile Robots | *Collaborative research*

Studied deep reinforcement-learning approaches that combine human guidance with autonomous control for mobile robot navigation, resulting in a peer-reviewed journal publication.

PUBLICATIONS AND PREPRINTS

Shaik, S., Smereka, J. M., & Wang, Y. (2026). *Multi-Agent Deep Reinforcement Learning Under Constrained Communications*. arXiv preprint arXiv:2601.17069.

Shaik, S., Parameshwaran, A., Nayak, A., Smereka, J. M., & Wang, Y. (2026). *MA-VLCM: A Vision Language Critic Model for Value Estimation of Policies in Multi-Agent Team Settings*. arXiv preprint arXiv:2603.15418.

Nayak, A., **Shaik, S.**, & Wang, Y. (2026). *Belief-Aware VLM Model for Human-like Reasoning*. arXiv preprint arXiv:2604.09686.

Shaik, S., Smereka, J. M., & Wang, Y. (2025). *Generalized Advantage Estimation for Distributional Policy Gradients*. Proceedings of the American Control Conference, 3073–3078.

Tian, C., **Shaik, S.**, & Wang, Y. (2021). *Deep Reinforcement Learning for Shared Control of Mobile Robots*. *IET Cyber-Systems and Robotics*, 3(4), 315–330.

RESEARCH SOFTWARE AND REPRODUCIBILITY

Open-source LGMA implementation. Maintains a public PyTorch research prototype with interchangeable LGMA and conventional multi-head attention, synthetic and TinyStories experiments, and VLA data/model/evaluation modules. [Repository](#)

Experiment integrity. Built unit and smoke tests, configuration-driven experiments, resumable checkpoints, validation workflows, per-GPU throughput and memory accounting, and automated summaries for validation-loss-versus-token and validation-loss-versus-GPU-hour comparisons.

Research-computing readiness. Prepared bf16 CUDA training, gradient accumulation, single- and multi-GPU DistributedDataParallel execution, Slurm launch patterns, dataset staging, and portable Conda/Apptainer environment plans for accelerator-based studies.

TECHNICAL SKILLS

Programming	Python; scientific computing and research software development
Deep Learning	PyTorch; transformer models; attention mechanisms; reinforcement learning; VLM/VLA
GPU & Scale	CUDA; bf16 mixed precision; PyTorch DistributedDataParallel; torchrun; Slurm
Research Stack	Hugging Face Datasets/Transformers; NumPy; h5py; OpenCV; Weights & Biases
Robotics & Tools	MuJoCo/LIBERO workflows; Git/GitHub; Conda; Apptainer; pytest